Theoretical heuristics in digital social research

Hjalmar Bang Carlsen

Abstract:
For researchers in the social sciences and humanities, the digital data revolution has primarily been about data—its scale, variety, and granularity. The theoretical opportunities presented by these new data sources, however, have often been backgrounded. Digital trace data are generated for purposes other than social research, and their reappropriation requires creative theorizing. This chapter aims to foreground this theoretical work. It critically examines current approaches to theory testing and theory generation and offers an critique and points towards a reformulation of the computational grounded theory framework—one that places active theorizing and interpretation at the center of digital social research.

Keywords: theorizing, computational grounded theory, abductive analysis, digital methods

For researchers in the Social sciences and Humanities the digital data revolution has mainly been about data. The scale of data, the variety of data, the granularity of data. The new theories that we can build and test with these new data sources have been somewhat backgrounded. To a large extent theory is out and data is in. SSH courses and even whole degrees are dedicated to improving students' data handling and coding skills. Programming languages such as R and Python are valued skills by the labor market and students alike. The focus on data skills is to a large extent necessary development. It is hard to do stuff with large-scale digital data without the ability to effectively handle and analyze data. Yet, without the ability to creatively and critically work with theories digital data adds little value. It is through a theoretical lens that digital data goes from being *bad found data* (data not produced by researchers, but by organizations for non-scientific purposes) to valuable knowledge.

If we want to be good at exploiting found digital traces(social media posts, mobile sensory data aso), we need to be good at seeing the theoretical potentials of data and testing their appropriateness. (Ruppert et al., 2013) argue, digital data captures *transactions* rather than *individual accounts*, *continuous* rather than *bundled* time, and allows for unprecedented *granularity*. (Salganik, 2019) similarly stresses digital data bigness which allows for the study of *heterogeneity* and the analysis for analyzing patterns of *rare events*. Digital data also contains large-scale open-ended text data at the level of populations. While the specific features of digital data might change over time or between platforms they are likely to have features different from that of traditional data sources. All these features can be exploited to make theoretical innovations within a certain literature. Yet, the path to these *theoretical innovations* can differ.

Two different research strategies have dominated the reappropriation of digital traces, one largely deductive, using digital data to test existing theories, and one large inductive using digital as a resource for generating theory. Both these approaches have certain problems which to some extent have to do with the nature of digital data, but both highlight important aspects of working with digital data. I'll start by discussing the deductive strategy, before moving on to the inductive strategies, I'll spend more time on the latter as it occupies a very central place in SSH.

Digital data as a resource for testing theories

[J]ust as the invention of the telescope revolutionized the study of the heavens, so too by rendering the unmeasurable measurable, the technological revolution in mobile, Web, and Internet communications has the potential to revolutionize our understanding of ourselves and how we interact (Watts, 2012)

Sociologist John Levi Martin argues that social science is filled with theories and one central goal of research is to eliminate some of them and keep the ones that do a good job of explaining social life (Martin, 2018). Yet, our sources of data greatly limit our ability to do so because they can't capture many aspects of our social science theories. These theories then live a life as ghosts, too intangible to do anything empirical with, and yet, still tangible enough to frighten empirical research now and again. As when interactionist critique survey researchers for generalizing for the particular social situation of the survey to the individuals attitude more generally (Jerolmack & Khan, 2014). Surveys are well-designed to capture certain opinions, values, and demographic variables, but are bad at capturing whole networks, situations, interactions, and behavior. Yet, many social science theories are about social interaction, social relations, communication, and behavior. These theories of social life have been hard to measure using traditional methods, with digital data this has arguably changed. When the unmeasurable becomes measurable, old untested theories can suddenly be mobilized to challenge or contribute to explaining important social processes(Golder & Macy, 2014; Latour et al., 2012). This is especially true for quantitative data collection methods, not as much so for ethnography which collects data on human interaction through direct in situ observation. The discourse of the *unmeasurable made measurable* through digital data can and should be met with healthy skepticism. Measurement here refers to quantitative measures and testing refers to quantitative statistical tests, but there are other ways of measuring and registering social life and other ways of testing theories.

While theory testing is opposed to theory generation, there are in many cases of theory testing a creative act of theorizing. Researchers have to interpret a given case as an instance of some theoretical idea, mobilizing a given theory within a new domain and more (Tavory & Timmermans, 2014). This is especially so when working with digital data. Answering either "What theories could I test using this data?" or "What digital data do I need to test this theory" both demand searching and creatively matching theory to data. This demands active theorizing. Abbott book *Methods of Discovery*

(Abbott, 2004) provides a rich repository of different heuristics that researchers can use to come up with new theoretical propositions. I'll touch upon the heuristic of *shifting levels* and *making an assumption* below.

In my research, I have benefitted from being somewhat opportunistic in the interpretation of digital trace data. I have used Facebook groups of social movement as a means to collect group interaction from social movement at scale, something which before was an impossible empirical task. Drawing on group interaction data we could *shift levels* from the organizational actors to the patterns of within-group interaction in explaining collective action. Digital data can also become very powerful by making a strong *assumption*. Golder & Macy assume that Twitter posts are indicative of the mood of individuals, and in doing so they suddenly have a huge cross-cultural dataset on individual sentiment at a highly granular level(Golder & Macy, 2011). This enables them to study the temporal patterns of sentiments, finding that individuals are more positive in the morning than the rest of the day. A finding that is robust across cultures. Using social media posts as indicative of sentiment is now standard, importantly it varies if authors take it to indicate an expressed sentiment towards some object or the emotional state of the individual more generally. While many, including myself, are very skeptical towards sentiment analysis, it has been a highly successful strand of digital social research, built on a strong assumption about what social media text data can measure.

Yet, creative theory testing has some major drawbacks if the digital data appropriateness is not seriously addressed. Digital data is qualitative in the sense that it is typically data about what people do or say at specific points in time. Hence, we have to take very seriously the varying practices that generate the data we have at hand. For instance, if we use Facebook groups to study social movement group interaction we need to make sure rank and file members interact in these groups and not just movement leaders posting, with no real interaction. They might start as forums for group coordination but end up as forums for mass communication. Similarly, with Golder and Macy sentiment analysis we might ask if their findings still hold true if we account for communicative conventions. There might be communication conventions that make positive words more likely in the morning than the evening. Is it more common to say "good morning" to followers than "good night"? While we do not know if this is true we do know that there are conventions to communication and that these might very well also have a temporal order, hence we can't ignore conventions when estimating something like sentiment.

A fundamental problem of being too focused on using digital data as a means to test existing theories, namely that digital data is a result of a digitalization of social practices that likely alter the practices (Marres, 2017), hence demands some degree of reconceptualization of our existing theories and that researchers are open to giving up on certain theories that prove a bad match. To sum up on this section, a too narrow focus on testing established theories risks a bad match between data and theory and furthermore misses the opportunity to generate new theories. This brings us to the research strategies more concerned with generating theory from data and thereby, the argument

goes, ensure a better fit between theory and data.

Generating theory from data and methods - computational grounded theory

The most central methodological tradition within data-driven theory generation in social sciences has been grounded theory. I'll first cover the central idea of grounded theory as formulated by Glaser and Strauss(Glaser & Strauss, 2009) and then move on to its computational variant.

GT aims to generate theory from data analysis, as opposed to generating theory from logical deduction from prior assumptions. The latter, according to Glaser and Strauss, leads to too many opportunistic uses of theory that do not fit the data (Glaser & Strauss, 2009, p. 6). As an alternative to theoretical deduction, Glaser and Strauss argue that theory should be "systematically worked out in relation to the data during the course of research. Generating a theory involves a process of research." (Glaser & Strauss, 2009, p. 6). This is key to grounded theory, namely that theorizing is derived from a process of systematic data analysis. The method used is the constant comparative method which analyzes each instance of a category with prior instances in order to get at its defining features, central properties and relations to other categories. Typically, the constant comparative method relies on open coding of instances working towards an integrated theory. In this manner, GT combines the systematic coding procedure of more traditional quantitative content analysis, typically aimed at theory testing, with qualitative work aimed at developing theory, the latter is typically done without any systematic data analysis (Glaser, 1965). Critics have argued that *systematic data analysis* has taken over and in a sense, hampered theory generation (Tavory & Timmermans, 2014). Will get back to this critique again because it is equally relevant for computational grounded theory.

With the emergence of large digital and textual data scholars have furthered grounded theory to incorporate computational methods. Large-text data have become an attractive empirical resource, especially as the domains of social life that produce textual traces have expanded well beyond merely the political and cultural elite. The scale of the data makes computational methods partly necessary for navigating and analyzing the corpus. Furthermore, advances in computational methods for analyzing text has given scholars a systematic and effective framework for analyzing categories in text.

A key reference for this approach is Laura Nelson's Computational Grounded theory, which details the process of discovering, refining and verifying patterns. (Nelson, 2017) most clearly articulates the methodological framework behind the use of unsupervised methods and qualitative inquiry, coining it *computational grounded theory*. She details three steps: pattern discovery, pattern refinement and pattern confirmation. Key to computational grounded theory is that patterns are located by the computational model and not the human coder, i.e, it is *computationally led* to ensure against the biased and constrained researcher. The researcher can then judge the usefulness of the categories returned

by the model given substantive interests and read paradigmatic text to gain an in-depth understanding of the category and refine the interpretation. Lastly, Nelsons argues, in line with many others, that pattern confirmation can be done by correlating the model's measure of the category with some other variables that indicate the same category. In its present form, computational grounded theory relies on the assumption that, 1) unsupervised models locates the relevant clusters of meaning within a corpus 2) the researcher can learn from minimal immersion, and lastly 3) the best way of validating a classification is through indirect validation. I'll address each step shortly and then discuss some of the pitfalls of this approach.

Discovery

When confronted with large text data, researchers have the challenge of discovering what categories are in the data and interpreting large amounts of text. A central claim in CGT is that this dual challenge can be solved by making discovery computationally led. According to Nelson, the computational model's task in *pattern discovery* is to reduce the messy, complicated text into a simpler and more interpretable list or network of words (Nelson, 2017, p. 7). This allows for relevant categories to emerge from the data that the researchers, due to their own preconceptions or the complexity of the text, had not considered. This prevents the researchers from introducing their own biases into the material. The simpler representations are then interpreted and categorized in a fashion similar to normal content analysis, but, Nelson argues, this process is "fully and immediately reproducible" (Nelson, 2017, p. 13), unlike human-coded text. In sum, computational *pattern discovery* that uses actual frequencies of the (co-)occurrence of words to locate ideas or concepts in the data constitutes, according to Nelson, a more "reproducible and scientifically valid grounded theory" (Nelson, 2017, p. 13). In CGT, the task of constant comparison from grounded theory aimed at discovering new categories from the data is done by an algorithm that takes word distribution across documents as input in order to detect patterns.

Interpretation and refinement

In the *pattern refinement* step, the researcher performs what Nelson calls a *computationally guided deep reading*. This step is supposed to 1) confirm the plausibility, 2) add interpretation and 3) possibly modify the patterns in order to provide a holistic reading. Thus, we move from mainly discovering categories to interpreting them. According to Nelson, there are two problems the computational guidance is supposed to solve: the natural limits to scaling deep reading and the biased nature of our reading. Both of these concerns are handled by algorithms that identify texts "that are representative of a particular theme" and that can "calculate the relative prevalence of that category" (Nelson, 2017, p. 24). A careful application of these algorithms enables the researcher to "read and interpret any amount of text without the burden of reading the full text"(Nelson, 2017, p. 26). Furthermore, both the researcher and the research community "can trust that when a quote is chosen as an example of something, it is not an outlier but is indeed representative of some theme in the text" (Nelson, 2017, p. 26). From reading the most prevalent words and documents, the researcher then labels the topic as being about something specific and, based on the reading of the paradigmatic

cases, writes an analysis of the topic. Here, saturation is ensured by the model that has 'read' the entire corpus and now presents the paradigmatic exemplar to human analysts. Instead of the human having to read through an extensive number of cases, as is common in grounded theory, the computer has done so. In other words, while human analysts read and interpret, it is the computer that leads the selection of documents, and guarantees the robustness of the pattern and the representativeness of the documents. This assumes that the model has correctly measured and delimited a topic and that the meaning of words used in a particular text match on to the meanings that the words take on in the other texts. Furthermore, it assumes that the reader is able to identify and analyze the categories without substantial exposure. But what does this provide us with theoretically? Although the theoretical output is seldom addressed directly, one way to see it is as data driven descriptive categories of one's field that enables a novel understanding.

Limitation of computational grounded theory

I'll draw heavily on arguments presented in Carlsen and Ralund (2022) in what follows. In this article and related work we have critically assessed the methodological assumptions behind the workflow of computational grounded theory and wider use of topic models in the social sciences. LDA topic models is an unsupervised model that locates topics in text. It has been one of the most used computational text analysis methods in the social science and humanities.

Authors have argued that there are several limitations to this framework for theory generation as well as empirical grounding. In Carlsen and Ralund(2023) we argue that the lack of clear distinction between a phase of risky discovery and measurement hampers both phases. The LDA topic model is both used for discovery and measurement. This limits the risks that can be taken in the theory generation phase. LDA model is known to produce meaningless topics and therefore topic models *"must find what we know is there. Ultimately, a topic model's trustworthiness must be determined by informed human judgments."* (Ramage et al., 2009, p. 4). In other words topic models can help find what we expect and recognize, but not something unexpected and strange. This is far away from the ambitious theory generation potential where computationally *"induced data structures can surprise, challenging presumptions or pre-existing theory, and lead the social analyst to abductively generate new theory by imagining what would be socially required for those patterns to exist*"(Evans & Aceves, 2016, p. 25). The apparent strange and meaningless is what is the source of theory generations. As Peirce argues novel theoretical propositions are made through abduction where a novel theoretical proposition is made in order to account for a surprising fact(for a Peircean inspired digital data methodology see Koed Madsen, 2024).

We argue that if we separate the phase of discovery of categories in text from their grounding and measurement then it sets us free to use algorithms purposefully to discover new and risky interpretations of data. Tavory and Timmerman argue *defamiliarization* is a technique that enables us to rethink our empirical observations, seeing them from a new and strange perspective. Karell and

Freedman (2019) argues that LDA topic models support *defamiliarization*. Importantly, while models such as the LDA topic might provoke one's understanding of corpus, they are not designed to do so. This in contrast to what Evans and Foster (2024) argue for, namely an *inverse supervised model* trained to surprise the human interpreter.

The second problem we raised in Carlsen and Ralund (2022) is that the procedure does not ensure that the researcher is a qualified interpreter of the material. To be a qualified interpreter a researcher needs to engage in intensive and extensive reading. In line with qualitative researchers such as Tavory and Timmerman 2014, we argue that the ability to recognize something as a surprising fact with potentially far-reaching theoretical consequences demands a qualified interpreter. There is a lot of irrelevant noise in digital data and few surprising facts. Being able to see these surprising facts both demands a certain cultivation of attention, but also a familiarity with the field site that enables one to see something as surprising in the first place. As is a well-known criterion to ethnographers, first one must be a competent actor, mastering the requirements to interaction in one's fieldsite, before any analytical claims are made (Lichterman 2017).

A third class of problems, we address in Carlsen and Ralund 2022, concerns the empirical robust results needed to support theory generation. The ambitious use of unsupervised methods such as LDA topic models greatly surpasses their ability and original design. When LDA topic models are used to provide an overview of the themes in the corpus, being an alternative to the systematic data analysis of the researcher in grounded theory, they assume that LDA captures all relevant themes. But LDA topic models are known to "hide" clusters within other clusters, hence any given topic returned by a model is likely to contain many different topics. Furthermore, the idea that topic models return topic and themes assumes a specific data generation process which in many cases does not hold. The reason for two words to co-occur is not necessarily that they come from the same theme or topic. In news-heavy discourse, news events are responsible for generating co-occurrences.

A final problem, according to Carlsen & Ralund (2022), is using top words and documents as representative of the entire topic. Words or even clusters of words do not mean anything, they are very uncertain indications of meaning, ensuring that one's interpretation is correct demands analyzing the entire sentence, document, or even communicative exchange. The reading of topic documents helps but ultimately does not solve the issue. The idea that a model can select the paradigmatic or representative document that a researcher can read and generalize to the entire topic is attractive, yet also one that demands a huge evidential burden before we should believe it. First of all, it is worth noting that a paradigmatic case and representative case are not the same, although both are used to describe the top document returned by a topic model. A paradigmatic case is the most clear and clean example of a given logic, in no other cases might the logic be formulated with such force and consequence, yet the paradigmatic case helps us recognize and analyze a certain logic to be able to detect it at work in other cases. A representative case represents a larger group of

empirical cases in an adequate way. While Biggie might be a paradigmatic rapper, most clearly explicating rap music as a genre of music, he is not representative of your average rapper. A topic model does not ensure that the top documents are either representative nor paradigmatic. They might be but we do not know. According to Carlsen and Ralund 2022 this is because word distributions are very uncertain indications of a certain meaning. Hence, if a document contains a lot of a certain themes and words this does not mean that it represents the rest of the documents nor does it mean that it most clearly captures its logic. That said it does make it more likely that the document is actually about the topic and not mainly about something else. According to Carlsen and Ralund it is not a bad reading strategy to read the top documents to begin with, but it is by no means enough to claim that one has a representative understanding of many thousands of other documents.

Some pointers towards a solution

While there are important critiques of both the deductive and inductive approaches to digital data, I think they have something very important to teach us in terms of theorizing with large-scale digital data. The deductive strategy stresses the importance of creatively *imagining* how the digital data might support testing or developing certain theories that before were hard due to data constraints. The inductive strategy focuses on *discovering* patterns within data that help us develop or question established theories. Then the researcher can move towards working out the details of the theoretical proposition in close dialog with the empirical data, purposefully *grounding* their interpretation. Once the categories have been properly grounded and the researcher has arrived at a robust understanding of a certain category the research can move towards *classification* and *measurement,* before the application of the variable in some statistical inference. In what is presented below I'll primarily talk about the 3 first steps: *imagining*, *discovering* and *grounding*. I'll to a large extent follow the process described by Tavory and Timmerman in their book on Abductive Analysis(2014). These points will be addressed in relation to text analysis, but they can be applied more broadly within digital social research.

Imagining

In the beginning of a research project it is useful to think about what theoretical moves a digital data source could support. To begin with this only demands limited data analysis, while more extensive theoretical engagement, finding out which theories might help with discovering a new angle on the research questions at hand. At this point of theory generation, it is fine to have many competing heuristics at play. The goal is to have a set of theoretical paths to follow that enable you to have something relevant to say to a community of researchers that are interested in the same as you.

To give a short example from my own research. We have collected a large Facebook dataset containing Danish political actors' posts and users' likes and comments on these posts. The first

question we ask ourselves is *what could this data be seen as a case of?* One obvious interpretation is political participation, when Facebook users like or comment on politicians they are engaging in politics in a similar fashion to writing letters to politicians or signing petitions. This operationalization would provide us with individual level, continuous, observational data on political participation for all citizens interacting with politicians on Facebook. Yet, this interpretation might be challenged with reference to lack of cost and the highly reactive and disorganized nature of liking and commenting on politicians posts. Liking is too cheap an action to have any resemblance to real participation, being what authors have called clicktivism, as Lewis et al 2014 state "Facebook conjured an illusion of activism rather than facilitating the real thing" (Lewis et al., 2014). An alternative interpretation would be to see users' likes and comments as indicative of the public opinion. After all, reacting to a politician's post is similar to reacting to a survey item in that it can be seen as an "opinion statement" constructed in response to an issue(Zaller, 1992). Researchers have attempted to use social media trace data as a low-cost substitute for survey data. Yet, due to the amongst other things that social media publics are not representative of national publics there has been both conceptual and empirical push-back to using social media data as data on the public opinion (Conrad et al. 2019). There are huge *selection problems*, both in regards to who is active and what is said on social media (Conrad et al., 2021). Thinking across both ideas a third option appears namely thinking about likes and comment to political elites as the *mobilized public opinion*. That is as the public that mobilizes their opinion on social media. This reconceptualization of public opinion makes it both more adequate in describing what is going on on social media and opens up a whole set of questions around who mobilizes their opinion, how and with what effects on political elites. In this way the *selection problem* becomes a source of theoretical innovation.

Discovery

The next step is to empirically investigate if the theories are applicable in one's case and how to further develop one's initial ideas. The idea of the mobilized public opinion, for example, assumes that Facebook users and politicians use Facebook to express their opinions. So we need to explore the data in order to find out what is going on. Carlsen and Ralund(2022) CALM(computer assisted learning and measurement) framework is a methodology for computational text analysis that tries to incorporate a researcher-led approach. We follow Laura Nelson's framework in using unsupervised machine learning methods to explore and discover topics within large text datasets. But the CALM framework uses the clusters returned by the model as an uncertain indication of categories. The top-words of relevant clusters are used to build search queries used for sampling documents for intensive and extensive reading. This query list is expanded using word embedding to locate similar words that might be of interest to a certain category. At this stage the researcher is engaging in discovery and hence the target is a list of words that enables the researcher to explore variation within a category and not a precise measurement of those categories. Therefore the query terms can return false positives, documents not about a certain category, as long as the amount of false positives does not become a problem for effective analysis of one's material. The query terms can continually be updated in light of the researcher changing interpretations. In the CALM framework

the returned clusters are not seen as exhaustive of the relevant categories in the data nor are the boundaries between different categories something that the research needs to follow, although they of course need to be considered.

Grounding

After having explored the corpus for finding theoretical perspectives that seem promising the phase of grounding. In this phase the list of themes or focuses is reduced to the most promising and the goal is to ground one's ideas more in the empirical material and work out the details of one's empirical perspective. In the case of the mobilized public opinion it could be to develop a typology of ways in which citizens mobilize their opinion and specify the relation between these different ways. Carlsen and Ralund 2022 propose the keyword document sampling procedure introduced above as the central ways to purposefully sample and compare instances and thereby properly ground and develop one's qualitative analysis of a category. Keywords are found using a fast word embedding model called word2vec which allows one to move effectively around the semantic space to try out one's interpretation in different contexts and to purposefully seek out boundary cases in order to specify the boundaries of one's category. We can then work towards having an empirically grounded conceptualization of the mobilized public opinion on Facebook. Describing both how the opinions are articulated and how they differ and propose a set of theoretical expectations to how they might influence the political elites, a set of propositions that can in turn be developed and tested quantitatively using text measures prepared for in the discovery and grounding phase.

Concluding remarks

While my examples throughout this paper have revolved around computational analyses of social media text data, the epistemological points have broader relevance for working with many types of digital data sources. Many digital data sources share with social media text data that they capture people in specific social situations which we traditionally have had a hard time measuring at scale. Think of data on people's searches online. This is an immense wealth of data whose significance for understanding of social life we have barely scratched the surface of. Although arguably useful for studying public opinion or epidemics, this is by no means exhaustive of the theoretical possibilities in search data. A more theoretical and curious engagement with search data could open up a highly productive avenue of research. Similarly with mobile sensory data that provide information of the movement and social interaction of people at unprecedented scale and granularity. The fact that we can measure who is together with whom in what social setting, provides a whole new set of questions bridging the gap between social interaction analysis and social network analysis. Again providing fruitful grounds of abductive inference. One important difference between working with text data and sensory data is the difficulty in reconstructing the actual interaction with sensory data. In other words when interaction is online it is captured and to a larger extent the researcher able to see what the interactants themselves have seen. Sensory data of physical movement and interaction

is typically much more limited and hence does not lend itself to the same type of qualitative analysis presented above. Nevertheless we can start to theorize and test what the size of interacting groups has consequences for networks, how interaction group size varies across socio-demographic groups, what social settings forster tie formation of ties and much more.

As a closing remark I'll quickly address the consequence of large language models(LLM) on the research process outlined here. In a sense LLM can and will be used throughout such a research process. You can get generative LLM to come up with theoretical concepts or propositions exploiting its vast and varied training data. You can quickly make it propose theories where it borrows concepts from another literature, making it try out different analogies which is an important part of theoretical discovery(Abbott 2004). While the soundness and creativity of these propositions is for the knowledgeable researcher to investigate and ultimately decide, generative LLM might prove to be an effective companion in the phase of imagining. On the other hand it might also lead to too much and too bad theorizing due to the reduced cost in formulating new theories. For discovery researchers have already used LLMs as alternative ways to cluster documents according to semantic similarity, summarize a large number of texts and more, and even having conversation with LLMs fine-tuned on a corpus. These methods might prove as more effective ways to discover novel insights as opposed to clusters of words. Both clusters of words and LLM representations should be treated as highly uncertain indications of meaning in the process of discovery, awaiting future grounding through intensive and extensive qualitative analysis. Critical to this research process is the ability of LLM powered approaches to effectively sample documents in ways that assist the human researcher fourth running theorizing of the empirical material. After all, word match is quick and cheap while LLM inference is costly, with non-trivial time and resource costs. While it still an open question how precisely LLM will influence working mixed methods with large text data, it is certain that it will be incorporated, I hope for my part that we keep it role in assisting rather than replacing human interpretative efforts.

References

Abbott, A. D. (2004). *Methods of discovery: Heuristics for the social sciences (contemporary societies)*. WW Norton & Company.

Carlsen, H. B., & Ralund, S. (2022). Computational grounded theory revisited: From computer-led to computer-assisted text analysis. *Big Data & Society*, *9*(1), 20539517221080146.

Conrad, F. G., Gagnon-Bartsch, J. A., Ferg, R. A., Schober, M. F., Pasek, J., & Hou, E. (2021). Social Media as an Alternative to Surveys of Opinions About the Economy. *Social Science Computer Review*, *39*(4), 489–508. https://doi.org/10.1177/0894439319875692

Evans, J. A., & Aceves, P. (2016). Machine Translation: Mining Text for Social Theory. *Annual Review of Sociology*, *42*(1), 21–50. https://doi.org/10.1146/annurev-soc-081715-074206

Evans, J. A., & Foster, J. G. (2024). Algorithmic Abduction: Robots for Alien Reading. *Critical Inquiry*, *50*(3), 375–401. https://doi.org/10.1086/728933

Glaser, B. G. (1965). The Constant Comparative Method of Qualitative Analysis. *Social Problems*, *12*(4), 436–445. https://doi.org/10.2307/798843

Glaser, B. G., & Strauss, A. L. (2009). *The discovery of grounded theory: Strategies for qualitative research* (4. paperback printing). Aldine.

Golder, S. A., & Macy, M. W. (2011). Diurnal and Seasonal Mood Vary with Work, Sleep, and Daylength Across Diverse Cultures. *Science*, *333*(6051), 1878–1881. https://doi.org/10.1126/science.1202775

Golder, S. A., & Macy, M. W. (2014). Digital Footprints: Opportunities and Challenges for Online Social Research. *Annual Review of Sociology*, *40*(1), 129–152. https://doi.org/10.1146/annurev-soc-071913-043145

Jerolmack, C., & Khan, S. (2014). Talk is cheap: Ethnography and the attitudinal fallacy. *Sociological Methods & Research*, *43*(2), 178–209.

Karell, D., & Freedman, M. (2019). Rhetorics of Radicalism. *American Sociological Review*, *84*(4), 726–753. https://doi.org/10.1177/0003122419859519

Koed Madsen, A. (2024). Digital methods as 'experimental a priori' – how to navigate vague empirical situations as an operationalist pragmatist. *Convergence*, *30*(1), 94–115. https://doi.org/10.1177/13548565221144260

Latour, B., Jensen, P., Venturini, T., Grauwin, S., & Boullier, D. (2012). The whole is always smaller than its parts: A digital test of Gabriel Tarde's monads. *British Journal of Sociology*, *63*(4), 590–615.

Lewis, K., Gray, K., & Meierhenrich, J. (2014). The structure of online activism. *Sociological Science*, *1*, 1–9.

Lichterman, P. (2017). Interpretive reflexivity in ethnography. *Ethnography, 18(1)*, 35-45.

Marres, N. (2017). *Digital sociology: The reinvention of social research*. John Wiley & Sons.

Martin, J. L. (2018). Thinking Through Statistics. University of Chicago Press. https://press.uchicago.edu/ucp/books/book/chicago/T/bo28394847.html

Nelson, L. K. (2017). Computational Grounded Theory: A Methodological Framework. Sociological Methods & Research, 004912411772970. construct validity. https://doi.org/10.1177/0049124117729703

Ramage, D., Rosen, E., Chuang, J., Manning, C. D., & McFarland, D. A. (2009). Topic modeling for the social sciences. NIPS 2009 Workshop on Applications for Topic Models: Text and Beyond, 5(27), 1–4. http://nlp.stanford.edu/dramage/papers/tmt-nips09.pdf

Ruppert, E., Law, J., & Savage, M. (2013). Reassembling Social Science Methods: The Challenge of Digital Devices. *Theory, Culture & Society*, *30*(4), 22–46. https://doi.org/10.1177/0263276413484941

Salganik, M. J. (2019). *Bit by bit: Social research in the digital age*. Princeton University Press.

Tavory, I., & Timmermans, S. (2014). *Abductive analysis: Theorizing qualitative research*. University of Chicago Press.

Watts, D. J. (2012). *Everything Is Obvious: How Common Sense Fails Us* (5/27/12 edition). Crown Currency.

Zaller, J. R. (1992). *The Nature and Origins of Mass Opinion*. Cambridge University Press. https://doi.org/10.1017/CBO9780511818691